



Rasika Edirisinghe

MSCA Doctoral Researcher

🔗 CASCADE MSCA Doctoral Network | University College Cork
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Summary

MSCA Doctoral Researcher at University College Cork, funded through the CASCADE Horizon Europe Marie Skłodowska-Curie Doctoral Network. My research focuses on the computational representation and retrieval of textual relations under transformation, particularly where lexical similarity is weakened or lost through paraphrase, adaptation, obfuscation, allusion, historical variation, and related forms of textual change.

My current work examines how such relations can be encoded, retrieved, ranked, and evaluated at scale across large textual collections. It includes semantic retrieval, parallel-text modelling, relation judgement, benchmark construction, and visual analytics for moving between corpus-level patterns and close reading.

Prior to my doctoral research, I worked for more than four years in data analytics and implementation engineering. From April to July 2026, I completed a research secondment at Nimbus Research Centre on grounding and trust in agentic cyber-physical systems within the inTrusted project.

Education

University College Cork

PhD in Computer Science, MSCA Doctoral Researcher

Ireland
Sep 2024 – Sep 2027 (expected)

Research focus: *Modelling Intertextual Relations under Transformation: Retrieval, Judgement, and Visual Analytics across the Parallel Text Spectrum*
CASCADE Horizon Europe MSCA Doctoral Network

- Developing computational representations for textual relations across paraphrase, adaptation, obfuscation, allusion, historical variation, and related forms of transformation.
- Investigating large-scale retrieval methods for related passages with weak or absent lexical overlap.
- Constructing benchmarks and evaluating relation judgements across different forms and degrees of transformation.
- Designing visual-analytics methods for exploring relation spaces and supporting close reading.

Robert Gordon University

M.Sc. Business Analytics, Distinction

Aberdeen, UK
Sep 2021 – Sep 2023

Thesis: *Twitter Sentiment Reason Mining Framework to Identify Major Problems in US Healthcare Industry*

Grade: A

Developed a Python-based NLP framework using sentiment analysis, topic modelling, clustering, and text vectorisation to analyse Twitter discourse on the US healthcare sector.

Cardiff Metropolitan University

B.Sc. (Hons) Business Information Systems

Cardiff, UK
Nov 2016 – Oct 2018

Thesis: *Computer-Aided Diagnosis System for Lung Cancer Detection* (Java-based CADe system)

Pearson Edexcel

BTEC Higher National Diploma, Software Development

UK
May 2014 – May 2016

OOP · Data Structures & Algorithms · Networking · Web Applications · Database Development

Academic Experience

University College Cork

Teaching Assistant & Co-Lecturer

Ireland
2025/26

Data Visualization for Machine Learning (CS3205 / CS6426)

- Revamped lecture and lab materials for the 2025/26 cohort.
- Delivered lectures and hands-on Python lab sessions.
- Produced all lecture slides for the 2026 delivery.
- Assessed and graded student assignments and examinations.

University of Ruhuna, Dept. of Electrical & Information Engineering

Visiting Lecturer

Galle, Sri Lanka
Aug 2023 – Sep 2024

- **Introduction to Data Visualization with Power BI** — DAX, dashboards, Gestalt principles, color theory.
- **Excel for Data Science** — PivotTables, time-series, statistical analysis, Solver.
- **Data Preprocessing & Cleaning / Data Splitting & Cross Validation**

Sri Lanka Telecom Training Centre, Sri Lanka Telecom PLC

Visiting Lecturer

Welisara, Sri Lanka
Aug 2023 – Sep 2024

Pearson BTEC delivery:

- **Fundamentals of AI & Intelligent Systems** (K/618/5660) — AI types, NLP, conversational AI, ethics.
- **Big Data & Visualization** (J/618/6251) — Python, Jupyter, Power BI, DAX, data governance.
- **Business Intelligence** (M/615/1641) — BI systems, OLAP, descriptive & predictive analytics.

Industry Experience

EFL Global

Data Analyst

Colombo, Sri Lanka (Hybrid)

May 2023 – Sep 2024

- Automated global finance reporting across 39 countries & 70+ offices (Python, VBA, Advanced Excel).
- Maintained Azure Databricks workloads supporting automated MIS, AR & AP reporting.
- Built Power BI & Tableau dashboards; implemented workflows via Power Automate & Logic Apps.
- Managed Azure Analysis Services, Data Lakes, Blob Storage, and Synapse Analytics.

Affno Asia Pacific

Implementation Engineer

Colombo, Sri Lanka

Nov 2022 – May 2023

- Designed and deployed BI solutions for supply chain analytics; acted as Technical Business Analyst.
- Provided training and support to clients; worked with Amazon EC2 & MongoDB Atlas.

SYNERGEN Health

Operations Analyst

Colombo, Sri Lanka

Sep 2019 – Nov 2022

- Managed Qlik-Sense cloud-based RCM BI platform (Doctrix) for real-time KPI & SLA monitoring.
- Led process training across multiple clients in Revenue Cycle Management.
- Implemented Python & VBA automation for financial reporting; maintained AWS DynamoDB & Kinesis.

SYNERGEN Health

Associate Operations Analyst (Intern)

Colombo, Sri Lanka

Mar 2019 – Sep 2019

Client reporting, process monitoring, data sanity checks, and trade support.

Global System Solutions

Business Analyst (Intern)

Colombo, Sri Lanka

Sep 2018 – Mar 2019

Requirements gathering, use case development, reporting specification, pre-sales support.

Publications & Conference Outputs

Jul 2025: Rewriting Tradition: Quantifying Change in Lady Gregory's Irish Legends

R. McCarthy, R. Edirisinghe, J. O'Sullivan, C. Ó Gallchoir, R. Minghim, Ó. Murphy

DH2025: Alliance of Digital Humanities Organizations, Lisbon, Portugal

doi.org/10.5281/zenodo.18340803

Published Abstract & Presentation

Jun 2023: Sentiment Reason Mining Framework for Analyzing Twitter Discourse on Critical Issues in US Healthcare Industry

R. Edirisinghe, P. P. G. Dinesh Asanka

2023 International Research Conference on Smart Computing and Systems Engineering (SCSE)

ieeexplore.ieee.org/document/10215010

Conference Abstracts & Presentations

Sep 2026: Evaluating Lexical Extraction in Hiberno-English: Rule-Based and LLM-Assisted Approaches

R. Edirisinghe

Digital Humanities Congress 2026, University of Sheffield

dhi.ac.uk/dhc/2026/paper/320

Conference Presentation

Jan 2026: Modeling Parallel Text: A Multidimensional Typology of Authorship and Transformation

R. Edirisinghe

CASCADE × MECANO Convention 1, 28 January 2026

CASCADE Convention 1 short abstracts

Published Abstract & Presentation

Current Research

- Modelling Intertextual Relations under Transformation** — representation, retrieval, relation judgement, benchmark construction, and evaluation across the parallel-text spectrum.
- Relation-Space Exploration and Visual Analytics for Parallel Text** — interactive exploration connecting corpus-scale patterns with close reading of individual textual relations.
- Trust and Grounding in Agentic Cyber-Physical Systems** — research conducted during the Nimbus secondment with Tharindu Ranathunga on language-to-action propagation through planning and tool use.

Talks & Academic Contributions

Sep 2026: Evaluating Lexical Extraction in Hiberno-English: Rule-Based and LLM-Assisted Approaches

Digital Humanities Congress 2026, University of Sheffield

Oct 2025: Beyond Translation: Rethinking 'Parallel Text' Across Disciplines

UCC Digital Humanities Colloquium, University College Cork

Apr 2025: Language Modelling and Visualization for Parallel Text

CACSSS PhD Lightning Talks, University College Cork

29 Jun 2023: Sentiment Reason Mining Framework ... (paper presentation)

2023 International Research Conference on Smart Computing and Systems Engineering (SCSE)

Feb 2025: Peer Reviewer — 8th International Research Conference on Smart Computing and Systems Engineering

Oct 2023: Evaluator — System Development Project Poster Presentation, University of Kelaniya

Aug 2023: Curriculum Reviewer — Foundation to Data Science, University of Ruhuna

Research Secondment

Nimbus Research Centre

Cork, Ireland

Research Secondment — inTrusted Project

Apr 2026 – Jul 2026

Completed research secondment on the TU RISE-funded inTrusted project. The work examined grounding and trust in agentic cyber-physical systems, focusing on how intent can be preserved or altered as natural-language instructions pass through conversational interpretation, planning, tool use, and robotic action.

Skills & Technologies

Languages: Python, SQL, Java, JavaScript, C#, R, VBA

NLP & Retrieval: Hugging Face Transformers, SentenceTransformers, spaCy, Gensim, NLTK, BM25, ANN search, LLM APIs

ML / DL: PyTorch, TensorFlow, Keras, scikit-learn, XGBoost

Visualisation: Power BI, Tableau, Qlik Sense, Plotly, Matplotlib, Streamlit, DAX

Cloud: Azure (Databricks, Synapse, Blob, Logic Apps), AWS (EC2, DynamoDB, Kinesis), MongoDB Atlas

Dev Tools: Git, Jupyter, Weka, Orange, HTML/CSS, ASP.NET, Power Automate, Bash

Soft Skills: Technical communication with non-technical audiences, tutoring & mentoring, business process design